

UPC INSTRUMENTS PRIVATE LIMITED

Calibration Procedure for Extractive Type Online Water Analyzer

Model: UPC-WA-202

Parameters: BOD, COD, TSS, pH

1. OBJECTIVE

To ensure the reliable and accurate measurement of water quality parameters—**BOD, COD, TSS, and pH**—through a two-point calibration method using standard reference solutions. Calibration is essential to maintain compliance with **CPCB norms** and operational accuracy.

2. CALIBRATION REQUIREMENTS

Instruments and Materials:

Item	Description
Standard Solutions	Certified standards for BOD, COD, TSS (e.g., GGA, Phthalate)
Buffer Solutions	pH 4.0, 7.0, and 9.2 buffers
Blank Solution	Deionized or distilled water
UPC Analyzer Touch Interface	For manual calibration entry
Clean Glass/Plastic Beakers	For holding and feeding solutions
PPE	Gloves, lab coat, goggles

3. CALIBRATION FREQUENCY

- Monthly (or as per operational SOP)
 - Post-maintenance or sensor replacement
 - Upon significant deviation in results
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4. CALIBRATION PROCEDURE (PARAMETER-WISE)

A. pH Calibration (Electrode-Based Method)

1. Navigate to **Calibration Tab (UI:5)** on the touch interface.
 2. Rinse probe with distilled water, then immerse in **pH 7.0 buffer**.
 3. Once stabilized, set this as the **Reference Point**.
 4. Rinse and then insert probe into **pH 4.0 or 9.2 buffer** (choose one as Span).
 5. Once stable, confirm as **Span Value**.
 6. Rinse and return to normal operation.
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B. COD Calibration (UV-Vis Method)

1. Access **Calibration Tab (UI:5)** on display.
 2. Pass **blank (distilled water)** through sample inlet.
 3. Let the analyzer stabilize and set it as **Zero Calibration**.
 4. Pass **standard phthalate solution** (e.g., 100 mg/L COD).
 5. Once stable, enter the known value and press OK to **set Span Calibration**.
 6. Rinse with distilled water.
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C. BOD Calibration (UV-Vis Method)

1. Follow same calibration UI as COD.
 2. Pass **distilled water** and set **Zero Point**.
 3. Then pass **GGA solution** (e.g., 200 mg/L BOD).
 4. Wait for stable reading and enter span value accordingly.
 5. Confirm and rinse the flow path.
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D. TSS Calibration (UV-Vis Method)

1. Insert **distilled water** sample first to set **Zero Calibration**.
2. Pass **Formazin standard** or certified turbidity/TSS solution.

3. Once the reading stabilizes, input the value as **Span Calibration**.
4. Save and rinse the chamber to complete calibration.

5. FLOW METER CALIBRATION (If Installed)

1. Navigate to **Flow Calibration** under **Settings Tab (UI:2)**.
2. Close pump, observe zero flow, and press **Zero**.
3. Start pump, wait until stable, enter flow value and press **Span**.
4. Flow calibration is now complete.

6. POST-CALIBRATION STEPS

- Reconnect sampling line to ETP sample.
- Allow analyzer to run 2–3 measurement cycles.
- Confirm stable, expected values are displayed.
- Record calibration in logbook or system memory.

7. DOCUMENTATION

Field	To be Filled
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Calibration Date	DD/MM/YYYY
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Technician Name	
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Parameters Calibrated	BOD, COD, TSS, pH
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Standard Values Used	e.g., 100 mg/L COD, pH 7.0
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Observations	
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Signature	
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8. SAFETY & NOTES

- Use **only fresh certified standard solutions**.
- Wear appropriate PPE during calibration.
- Rinse sensors after each calibration step.

- For accurate results, maintain clean flow cells and dosing channels.
- Contact **UPC Technical Support** if calibration fails or error persists.

Email: info@unitechmeter.com

Technical Support: +91 70822 00823

Website: www.unitechmeter.com

UPC INSTRUMENTS PRIVATE LIMITED

MFG Unit : Plot No. 125 Near Tehsil, Barsat Road Gharaunda-132114 (Karnal)
info@unitechmeter.com
